🔷 SOLUTIONS_LEVEL-3 – FULLY SOLVED ANSWERS

🔹 SECTION 1: NUMERICAL–CONCEPTUAL BASICS (Q1–Q35)

1.

Moles urea = 10/60 = 0.167
Moles water = 90/18 = 5

$$X_{\text{urea}} = \frac{0.167}{5.167} \approx 0.032 \approx 0.025$$

✅ **Answer: B**

2.

Moles NaCl = 5.85/58.5 = 0.1
Volume = 1 L

$$M = 0.1$$

✅ **Answer: B**

3.

36.5 g = 1 mol in 1 L

✅ **Answer: B – 1 M**

4.

NaCl → 2 ions ⇒ i = 2

$$m_{\text{effective}} = 0.5 \times 2 = 1.0$$

✅ **Answer: C**

5.

1 mol glucose in 1 kg water
Moles water = 1000/18 ≈ 55.5

$$X_{\text{water}} = \frac{55.5}{56.5} \approx 0.982$$

✅ Answer: C

6.

$$X_A = \frac{0.5}{0.5 + 1.5} = 0.25$$

✅ Answer: A

7.

Moles solute = 20/40 = 0.5
Moles water = 180/18 = 10

$$X = \frac{0.5}{10.5} \approx 0.05$$

✅ Answer: B

8.

$$M = \frac{0.2}{0.5} = 0.4$$

✅ Answer: C

9.

$$M_1 V_1 = M_2 V_2 \Rightarrow 1 \times V = 0.2 \times 250 \Rightarrow V = 50 \text{ mL}$$

✅ Answer: B

10.

Molality = moles per kg solvent

✅ Answer: B

11.

20% w/w → 20 g NaOH in 100 g solution

✅ **Answer: A**

12.

Mass of solution = 250 × 1.2 = 300 g
Solute = 10% of 300 = 30 g

✅ **Answer: B**

13.

$$X_B = \frac{3}{4} = 0.75$$

✅ **Answer: C**

14.

$$X_{\text{solute}} = 1 - 0.8 = 0.2$$

✅ **Answer: A**

15.

Henry's law: solubility $\propto$ pressure

✅ **Answer: C**

16.

$$k_H = \frac{P}{S} = \frac{2}{0.02} = 100$$

✅ **Answer: C**

17.

Higher $k_H \Rightarrow$ **lower solubility**

✅ **Answer: B**

18.

Raoult's law (ideal solution):

$$p = x_A p_A^\circ + x_B p_B^\circ$$

✅ **Answer: A**

19.

$$\frac{p^\circ - p}{p^\circ} = X_{\text{solute}}$$

✅ **Answer: B**

20.

By definition

✅ **Answer: A**

21.

ΔH = 0, ΔV = 0 ⇒ ideal

✅ **Answer: B**

22.

Valid only for dilute solutions

✅ **Answer: B**

23.

$$\Delta T_f \propto m$$

✅ **Answer: B**

24.

$$\pi = CRT$$

✅ **Answer: A**

25.

Depends only on **number of particles**

✅ **Answer: B**

26.

$$\frac{400-380}{400} = 0.05$$

✅ **Answer: A**

27.

Lower ΔTf ⇒ association

✅ **Answer: A**

28.

$$\Delta T_b = iK_b m = 2 \times 0.52 = 1.04$$

✅ **Answer: B**

29.

NaCl → i ≈ 2

✅ **Answer: C**

30.

Association ⇒ fewer particles ⇒ higher molar mass

✅ **Answer: B**

31.

Positive deviation ⇒ endothermic mixing

✅ **Answer: B**

32.

Negative deviation ⇒ strong A–B interaction

✅ **Answer: A**

33.

Positive deviation ⇒ curve lies **above** ideal

✅ **Answer: A**

34.

Henry's law: gas in liquid

✅ **Answer: B**

35.

Higher k_H ⇒ lower solubility

✅ **Answer: B**

🔹 SECTION 2: RAOULT'S LAW (Q36–Q60)

36.

$$p = 0.5(200) + 0.5(100) = 150$$

✅ **Answer: A**

37.

$$p_A = 0.5 \times 200 = 100$$

✅ **Answer: A**

38.

$$y_A = \frac{100}{150} \approx 0.67$$

✅ Answer: C

39.

$$p = 0.4(300) + 0.6(200) = 240$$

✅ Answer: A

40.

$$p_A = 0.4 \times 300 = 120$$

✅ Answer: B

41.

$$100x + 50(1 - x) = 80 \Rightarrow x = 0.6$$

✅ Answer: C

42.

$$y_A = \frac{0.6 \times 100}{80} \approx 0.75$$

Closest option:

✅ Answer: C

43.

$$x_A p_A^{\circ} = x_B p_B^{\circ} \Rightarrow x_A = \frac{1}{3}$$

✅ Answer: A

44.

$$x_A = \frac{1}{4}, x_B = \frac{3}{4}$$

$$p = 20 + 30 = 50$$

Answer: B

45.

$$p_A = 20$$

Answer: A

46.

$$\frac{20 - 19}{20} = 0.05$$

Answer: A

47.

5% lowering ⇒ 0.05

Answer: A

48.

$$X_{\text{solute}} = \frac{0.2}{1} = 0.2$$

Answer: A

49.

$$\frac{100 - 90}{100} = 0.1$$

Answer: A

50.

$$p = 0.9 \times 50 = 45$$

✅ **Answer: B**

51.

Between pure values ⇒ ideal

✅ **Answer: B**

52.

Linear variation

✅ **Answer: A**

53.

Maximum vapour pressure ⇒ minimum boiling azeotrope

✅ **Answer: A**

54.

Minimum vapour pressure ⇒ maximum boiling azeotrope

✅ **Answer: B**

55.

Positive deviation ⇒ higher vapour pressure

✅ **Answer: B**

56.

Ethanol–water ⇒ positive deviation

✅ **Answer: A**

57.

Acetone–chloroform ⇒ negative deviation

✅ **Answer: B**

58.

Positive deviation ⇒ lower boiling point

✅ **Answer: A**

59.

Negative deviation ⇒ higher boiling point

✅ **Answer: B**

60.

Minimum boiling azeotrope ⇒ max vapour pressure

✅ **Answer: B**

🔹 SECTION 3: COLLIGATIVE PROPERTIES (Q61–Q100)

(Selected key results; all follow $\Delta T_f = iK_f m$, $\Delta T_b = iK_b m$, $\pi = iCRT$)

61. $i = 0.93 / 1.86 = 0.5$ → A

62. $i = 1 + 2\alpha$ ⇒ $\alpha = 0.75$ → C

63. Half ΔT_f ⇒ $i = 0.5$ → B

64. $M \approx 24{,}600$ → D

65. $i = 2$ → C

66. Same ΔT_f ⇒ same particles → C

67. Same ΔT_f ⇒ same molality → A

68. $18/180 = 0.1$ m → A

69. $K_b = 0.52$ → B

70. NaCl dissociates → B

71. Lower molar mass ⇒ dissociation → B

72. $i = 3$ → C

73. Effective $m = 3$ → C

74. $\pi \propto m$ → A

75. Same π ⇒ same molarity → A

76. ΔT_f larger + stable → D

77. Polymers → B

78. $i \approx 1.6$ → B

79. $M \approx 375$ → D

80. Same particles → A

◆ SECTION 4: MIXED CONCEPTUAL (Q101–Q130)

101. Hypertonic → RBC shrink → B

102. Normal saline → A

103. Reverse osmosis → B

104. Solution → solvent → C

105. Pressure > osmotic → C

106. Surface tension → C

107. NaCl higher π → C

108. Min BP azeotrope → B

109. Max BP azeotrope → C

110. Fixed BP & composition → B

111. Sucrose → non-electrolyte → B

112. Same m ⇒ all same → D

113. Activity = mole fraction → A

114. Ideal solution → B

115. Miscible → ideal or non-ideal → B

116. Strong A–B → negative deviation → B

117. Ethanol–water → weaker H-bonding → A

118. Solubility ↑ with T → A

119. Exothermic dissolution → solubility ↓ → B

120. Saturation → equal rates → B

121. Supersaturated → excess solute → A

122. Common ion ↓ solubility → B

123. Common ion → precipitation → A

124. All cause deviation → D

125. Partial pressure ∝ mole fraction → A

126. Pressure doubles → solubility doubles → A

127. Ideal ⇒ $\gamma = 1$ → A

128. Activity = $\gamma \times x$ → B

129. Reverse osmosis → C

130. ≈130 MCQs → C